Alessandro Candeloro

Postdoctoral researcher, Università degli Studi di Palermo

Personal Data

30th of March, 1995, Bologna, Italy

Università degli Studi di Palermo
Via Archirafi 36, I-90123 Palermo, Italy

+393339174778

alessandro.candeloro@unipa.it

0000-0003-0582-4941

Google Scholar profile

Publons profile



Resume

I am a quantum physicist with broad interests in quantum information theory, with a focus on metrology and parameter estimation, quantum thermodynamics and quantum feedback. Measurement has been the central thread connecting most of my research: starting from the investigation of optimal schemes and ultimate bounds in quantum metrology, I later explored the role of measurement incompatibility in multi-parameter quantum estimation, addressing both foundational and applied aspects. I have developed novel methods that employ quantum measurements to implement feedback in continuous time quantum walks, enabling their controlled evolution toward specific targets. At a more fundamental level, I have examined how the third law of thermodynamics constrains the intersubjectivity of quantum outcomes, deriving fundamental limits to intersubjectivity in the quantum domains. I have also investigated the interplay between precision and dissipation in quantum systems coupled to multiple baths, uncovering how correlations influence fluctuations of quantum observables and thermodynamic irreversibility. My current research focuses on how measurement and thermodynamic determine the precision with which information can be acquired. Our final goal is to quantify the energetic resources required by measurement-based protocols, particularly those relevant to quantum computation. During my scientific career, I have acquired expertise that span from theoretical modelling of noisy quantum environments to programming and simulation for quantum trajectories.

Work experience

Sep. '26 → Sep '27	**Adjunct Professor**, School of Engineering, University of Palermo, *Palermo*, Italy.
	Physics 1 for the degree program in **Industrial and Information Engineering** (in English).
Jan. '25 → Apr '26	**Postdoctoral researcher**, University of Palermo, *Palermo*, Italy. Supervisor: Mauro Paternostro
Project	**Energetics of measurement-based quantum computation**
Jan. '23 → Dec '24	**Research Fellow**, Trinity College Dublin, *Dublin*, Ireland. Supervisor: Felix Binder
Project	**Emergence Of Objective Reality: From Qubit To Oscilloscope**

Education

Oct. '19 → Dec. '22	**PhD**, Università degli Studi di Milano, *Milan*, Italy. Supervisor: Matteo G.A. Paris
Thesis	**Characterization and engineering of quantum systems for quantum technologies**
Sep. '21 → Mar. '22	**Visiting Scholar**, Insititue Néel (CNRS), *Grenoble*, France. Supervisor: Alexia Auffèves
Project	**Two-body quantum energetics, fundamental and applications**

Sep. '17 → Oct. '19	**Master Degree in Theoretical Physics**, Università di Bologna, *Bologna*, Italy. Supervisor: Cristian Degli Esposti Boschi and Matteo G.A. Paris.
Finale grade	110/110 *with honours*. GPA: $29.75/30$

Sep. '14 → Sep. '17 **Bachelor Degree in Physics**, Università di Bologna, *Bologna*, Italy. Supervisor: Alexandr Kamenchtchik.

Final grade 110/110 *with honours*. GPA: 29.16/30

Publications

Preprint

1 E. Schwarzhans, A. Candeloro, F. C. Binder, M.P.E. Lock, P. Bakhshinezhad, *Average Equilibration Time for Gaussian Unitary Ensemble Hamiltonians*, arXiv:2603.28587 (2026)

Published

11 A. Candeloro, T. Debarba F. C. Binder, *Thermodynamic Constraints on the Emergence of Intersubjectivity in Quantum Systems*, Phys. Rev. A **113**(3), 032201 (2026)

10 S. V. Moreira, M. Radaelli, A. Candeloro, F. C. Binder, M. T. Mitchison, *Precision bounds for multiple currents in open quantum systems*, Phys. Rev. E **111**(1), 064107 (2025)

9 A. Candeloro, Z. Pazhotan, and M. G. A Paris, *Dimension matters: precision and incompatibility in multi-parameter quantum estimation models*, Quantum Science and Technology **9** 045045 (2024)

8 A. Candeloro, C. Benedetti, M. G. Genoni, and M. G. A Paris, *Feedback-assisted quantum search by continuous-time quantum walks*, Advanced Quantum Technologies, 2200093 (2023)

7 A. Candeloro, L. Razzoli, P. Bordone, and M. G. A Paris, *Role of topology in determining the precision of a finite thermometer*, Phys. Rev. E **104**(1), 014136 (2021)

6 A. Candeloro, S. Razavian, M. Piccolini, B. Teklu, S. Olivares, and M. G. A Paris, *Quantum probes for the characterization of nonlinear media*, Entropy **23**(10), 1353 (2021)

5 A. Candeloro, M. G. A Paris, and M. G. Genoni, *On the properties of the asymptotic incompatibility measure in multiparameter quantum estimation*, Journal of Physics A: Mathematical and Theoretical **54**(48), 485301 (2021)

4 A. Candeloro, C. Mereghetti, B. Palano, S. Cialdi, M. G. A Paris, and S. Olivares, *An enhanced photonic quantum finite automaton*, Applied Sciences **11**(18), 8768 (2021)

3 A. Candeloro, and M. G. A Paris, *Discrimination of ohmic thermal baths by quantum dephasing probes*, Phys. Rev. A **103**(1), 012217 (2021)

2 A. Candeloro, L. Razzoli, S. Cavazzoni, P. Bordone, and M. G. A Paris, *Continuous-time quantum walks in the presence of a quadratic perturbation*, Phys. Rev. A **102**(4), 042214 (2020)

1 A. Candeloro, C. D. E. Boschi, and M. G. A Paris, *Quantum probes for universal gravity corrections*, Phys. Rev. D **102**(5), 056012 (2020)

Invited talks and seminars

June 2025 **Second QWEST Workshop** Università degli Studi di Milan**,** *Milan, Italy*
TITLE: *Feedback-assisted quantum search by continuous-time quantum walks*

March 2025 **Quantum Theory Group in Palermo** Università degli Studi di Palermo, *Palermo, Italy*
TITLE: *Intersubjectivity with finite thermodynamical resources*

June 2024 **Quantum dynamics & control, Koch group** Freie Universität Berlin, *Berlin, Germany*
TITLE: *Feedback-assisted quantum search by continuous-time quantum walks*

Talks and posters

September 2026 **Quantum Roundabout 2026** The University of Nottingham, *Nottingham, England*
CONTRIBUTED TALK: *Parameters competing for precision: incompatibility in multiparameter quantum metrology*

June 2026 **Midsummer Conference on Open Quantum Systems, Decoherence, and the Quantum-to-Classical Crossover** University of Helsinki, Aalto University, *Helsinki, Finland*
CONTRIBUTED TALK: *Thermodynamic Constraints on the Emergence of Intersubjectivity in Quantum Systems*

September 2025 **17th Italian Quantum Information Science Conference** Università degli Studi di Bologna,
POSTER: *Precision bounds for multiple currents in open quantum systems* *Bologna, Italy*

July 2025 **Advances in Quantum Control: Techniques, Applications and Challenges (AQuTe25)**
Max Planck Institute for the Physics of Complex Systems *Dresden, Germany*
POSTER: *Feedback-assisted quantum search by continuous-time quantum walks*

June 2025 **Second QWEST Workshop** Università degli Studi di Milano, *Milan, Italy*
INVITED TALK: *Feedback-assisted quantum search by continuous-time quantum walks*

October 2024 **Quantum Information Days** Center for Theoretical Physics PAS, *Warsaw, Poland*
CONTRIBUTED TALK: *Intersubjectivity with finite resources*

September 2024 **Vienna Quantum Foundations Conference** IQOQI Vienna, *Vienna, Austria*
FLASH TALK AND POSTER: *Thermodynamics of broadcasting and the emergence of intersubjectivity*

June 2024 **Quantum Information and Probability (QIP24)** Linnaeus University, *Vaxjo, Sweden*
CONTRIBUTED TALK: *Intersubjectivity with finite resources*

September 2023 **Quantum Morocco** Faculty of Sciences at Mohammed V University, *Rabat, Morocco*
POSTER: *Feedback-assisted quantum search by continuous-time quantum walks*

June 2023 **Quantum Algorithms, Machine Learning and Control (QUAMCON 2023)**
Trinity College Dublin School of Engineering *Dublin, Ireland*
CONTRIBUTED TALK: *A photonic realisation of a quantum finite automaton*

June 2023 **Quantum Information and Probability (QIP23)** Linnaeus University, *Vaxjo, Sweden*
POSTER: *Bipartite quantum energetics and the quantum heat*

Semptember 2022 **Quantum characterization and control of quantum complex systems**
Fondazione Alessandro Volta and Lake Como School of Advanced Studies, *Como, Italy*
POSTER: *Feedback-assisted quantum search by continuous-time quantum walks*

August 2022 **Qalypso Summer School** University of Malta, *Gozo, Malta*
FLASH TALK AND POSTER: *Energetics of bipartite interacting quantum systems and quantum heat*

June 2022 **10th Quantum Thermodynamics Conference 2022** Queen's University Belfast, *Online*
FLASH TALK AND POSTER: *Energetics of bipartite interacting quantum systems and quantum heat*

October 2021 **13th Italian Quantum Information Science Conference (IQIS)** INFN, *Naples, Italy*
CONTRIBUTED TALK: *On the properties of the asymptotic incompatibility measure in multi-parameter quantum estimation*

March 2021 **Thermodynamics of quantum systems and processes** *Online*
Fondazione Alessandro Volta and *Lake Como School of Advanced Studies*
CONTRIBUTED TALK: *Discrimination of Ohmic thermal baths by quantum dephasing probes*

September 2020 **Young IQIS 2020 - Young Italian Quantum Information Science Conference** INFN, *Online*
CONTRIBUTED TALK: *Discrimination of Ohmic thermal baths by quantum dephasing probes*

Conference Organizer

July 2024 **Emergence of Classicality: New Perspectives on Measurements in Quantum Theory**
Trinity College Dublin: Organizing committee *Dublin, Ireland*

Schools

Semptember 2022 **Quantum characterization and control of quantum complex systems** *Como, Italy*
Fondazione Alessandro Volta and Lake Como School of Advanced Studies

August 2022 **Qalypso Summer School on Quantum Computing and Open Quantum Systems**
University of Malta *Gozo, Malta*

June 2022 **Solstice of Foundations** *Zurich, Switzerland*
ETH Zurich and *Squid*

March 2021 **Thermodynamics of quantum systems and processes** *Online*
Fondazione Alessandro Volta and Lake Como School of Advanced Studies

November 2020 **A mini-course on Quantum-Information Thermodynamics** *Online*
University of São Paulo

Teaching activities

Tutor and teaching assistant

A.Y. 2024/2025 **Advanced Open Quantum Systems and Quantum Information Theory course**
A.Y. 2023/2024 **Advanced Topics in Quantum Information**
A.Y. 2023/2024 **Python Project Module**, Bachelor's degree in Physics
A.Y. 2021/2022 **General Physics**, Bachelor's degree in Chemistry, held by Prof. Matteo G.A. Paris
A.Y. 2020/2021 **General Physics**, Bachelor's degree in Chemistry, held by Prof. Matteo G.A. Paris
A.Y. 2019/2020 **Thermodynamics**, Bachelor's degree in Physics, held by Prof. Claudio Grillo
Main activities Teaching, exercises, teaching support, student reception, written exams

Cosupervision of students

A.Y. 2024 **Srivathsan Srinivasan**, Master thesis *Evaluating the Distance from Quantum Objective States*
A.Y. 2024 **Sean Alexander**, Summer project *Strong Quantum Darwinism and conservation of energy*
A.Y. 2022 **Lorenzo Lanzini**, Bachelor thesis *Quantum Probes For Intrinsic Decoherence*
A.Y. 2021 **Gabriele Bressanini**, Master thesis *Multi-parameter quantum estimation in Gaussian systems*
A.Y. 2021 **Mattia Algranati**, Bachelor thesis *Thermally driven engine based on a single atom*

Science dissemination and other experiences

Apr. 2022 **Exhibition guide at the Italian Quantum Weeks** *Milano*
2020 → 2022 **Student representative in the Department Council** *Università degli Studi di Milano*
2020 → 2022 **Student representative in the PhD Council** *Università degli Studi di Milano*

Certificates

Ongoing **FOR24,** *Teaching certificate for High School*
24 CFU in anthropo-psycho-pedagogical disciplines and in teaching methodologies and technologies
August 2019 **IELTS Academic,** Overall Band Score: 7.0

Scientific activities

Referee Physical Review X, Physical Review Letters, Physical Review X Quantum, Quantum, Physical Review A, Physics Letters A, The European Physical Journal Plus, Scientific Reports, Indian Journal of Physics, Science Advances
Peer analysis Participant for the FQxI Metascience Project Review System
Participant for the Metascience Novelty Indicators Challenge

Scholarships and awards

A.Y. 2021/2022 **Erasmus + Traineeship**, Università degli Studi di Milano
Host institution: Institute Néel - CNRS. Duration: 6 months
A.Y. 2016/2017 **Study award in favor of deserving students**, Università di Bologna
A.Y. 2015/2016 **Study award in favor of deserving students**, Università di Bologna

Computer skills

Basic: LabView, Root; **Intermediate**: Matlab, C++, Excel, Word, PowerPoint, MacOs, Windows, Linux, Python, Julia; **Advanced**: Mathematica, LaTeX.

Languages

Italian **Mothertongue**
English **Independent user**
French **Basic user**

Other interests

I have always been interested in philosophical topics, especially epistemology and the philosophy of science. For these reasons, I attended a course from the Master's degree program in Philosophy entitled Philosophy of Science held by Raffaella Campaner. The course covered the conceptions of scientific explanation and causation in contemporary epistemology.
During my free time, I like reading, going to the cinema, and traveling. I enjoy cooking and experimenting in the kitchen (a strange thing for a theoretician, although it turns out to be the typical Italian stereotype). I am used to swimming to relax and recover mental energies: *mens sana in corpore sano*.